

CRF State Summary v19

Session: Generator Hierarchy, K35 Uniqueness,
 $\varepsilon_0 \leftrightarrow w_{\text{DE}}$, and Zero Free Structural Parameters

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Session summary. This session took the outside view further and entered the inside structure. Four key findings emerged:

- (1) K35 form identity is proved unique: $(d_s, b, n) = (3, 2, 8)$ is the only solution, which also proves Axiom 0 (δ is binary) retroactively.
- (2) $n = d_s + n_{\text{m}}$: the powerset of $\{x, y, z\}$ splits into singletons (space) and non-singletons (matter), with Hodge duality between space and gauge sector.
- (3) $\beta = d_s \mathcal{G}$ (gap 0.0006%): exchange rate is d_s copies of universal coupling; P-growth slows by exactly d_s at space crystallisation.
- (4) **Generator hierarchy**: CRF has zero free structural parameters (from form theorem) and exactly one free quantitative parameter (ε_0), which is observationally identical to w_{DE} (dark energy equation of state).

Key Numbers

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8$	$d_s = 3$	$\beta = 2.2121$	$F = 1.5377$	$T_{\text{avg}} = 35.57$
$\varphi = 1.61803$	$e = 2.71828$	$\pi = 3.14159$	$m = \varphi$	$m(m - 1) = 1$
$\mathcal{G} = 0.73737$	$\beta_{\text{pre}} = \mathcal{G}$	$f_{\text{II,crit}} = 0.14700$	$n_{\text{m}} = 5$	$n = d_s + n_{\text{m}}$

The Generator Hierarchy

Level 0 — Logic

δ is a binary distinction. *Not assumed: proved by K35 form uniqueness.*

Level 1 — Form theorem

$\sum_{k=0}^{d_s} \binom{d_s}{k} k (b^{d_s} - k) = b^{d_s} (b^{d_s} + 1)$ has unique integer solution $(d_s, b, n) = (3, 2, 8)$.

Consequences: $n_m = 5$, powerset split, 5 matter types $(1 + 3 + 1)$, Hodge duality.

Level 2 — Information theory

$\alpha = \ln 2$ (one bit). $D_0 = n(1 - e^{-1/n})$ (vacuum scale).

$\mathcal{G} = \alpha/D_0$ (coupling). $K^* = D_0/n$ (threshold). $\beta = d_s \mathcal{G}$ (rate).

Level 3 — The one free parameter

$\varepsilon_0 = 0.00755$ from EIC/SINDy dynamics.

Equivalent: $w_{DE} = -1 + \mathcal{G}/d_s = -0.754$ (DESI 2024, gap 0.03%).

ε_0 and w_{DE} are **the same degree of freedom**.

Level 4 — Physics

c, H_0 (measured). $G = \mathcal{G}c^2$. $|\dot{G}/G| = \mathcal{G}H_0$. $w_{DE} = -1 + \mathcal{G}/d_s$.

Zero structural free parameters. The structure $(d_s=3, b=2, n=8, n_m=5, \text{Hodge duality, 5 matter types})$ follows from the form theorem alone — no empirical input.

One quantitative free parameter. Given ε_0 (or equivalently w_{DE}), all coupling constants follow.

New Results This Session

K35 uniqueness theorem (algebraically proved)

K35 uniqueness

The form consistency condition

$$\sum_{k=0}^{d_s} \binom{d_s}{k} k (b^{d_s} - k) = b^{d_s} (b^{d_s} + 1)$$

has unique integer solution $(d_s, b, n) = (3, 2, 8)$.

Proof: For $d_s = 3$, this reduces to $b^6 - 11b^3 + 24 = 0$, giving $b^3 \in \{8, 3\}$. Only $b^3 = 8$ yields integer $b = 2$. For $d_s \neq 3$: no integer b exists.

Derives simultaneously: $d_s = 3$ (dimension), $b = 2$ (Axiom 0), $n = 8$ (modes).

$n = d_s + n_m$ and the powerset origin of matter

The $n = 8$ modes split as singletons $\{x\}, \{y\}, \{z\}$ (space) and non-singletons $\emptyset, \{xy\}, \{xz\}, \{yz\}, \{xyz\}$ (matter):

$$1_{\text{scalar}} + 3_{\text{gauge (2-forms)}} + 1_{\text{pseudo}} = 5 = n_m$$

Hodge duality: $\star\{x\} = \{yz\}$, etc. — space and gauge sector are dual.

Loop correction to Bethe percolation: $1/((n-1)n_m) = 1/35 = 2.857\%$ (actual 2.90%, gap 1.6%).

$\beta = d_s \mathcal{G}$ (gap 0.0006%)

$$\beta = d_s \mathcal{G} = \frac{3 \ln 2}{D_0} = 2.21211$$

Beta spectrum: $\mathcal{G}$ (pre-geometric), $2\mathcal{G}$ (Kerr), $3\mathcal{G}$ (current universe). CRF inflation endpoint: P-growth per event slows by exactly $d_s = 3$ at space crystallisation.

$\varepsilon_0 \leftrightarrow w_{\text{DE}}$: same degree of freedom

From CRF Paper v6: $w_{\text{DE}} = -1 + \mathcal{G}/d_s$. From K35: $\mathcal{G} = \sqrt{n(n+1)\varepsilon_0}$. Combined:

$$\varepsilon_0 = \frac{d_s^2(1 + w_{\text{DE}})^2}{n(n+1)} \quad \Longleftrightarrow \quad w_{\text{DE}} = -1 + \frac{\sqrt{n(n+1)\varepsilon_0}}{d_s}$$

DESI 2024: $w_{\text{DE}} = -0.754$ gives $\varepsilon_0 = 0.007565$ (CRF: 0.00755, gap 0.19%). The instability floor ε_0 and the dark energy equation of state w_{DE} are two observational windows on the same underlying quantity.

Candidate identity: $\alpha\sqrt{\varphi} \approx D_0^2$ (gap 0.22%)

$$\alpha\sqrt{\varphi} = \ln 2 \cdot \sqrt{\varphi} = 0.8817 \quad D_0^2 = 0.8836 \quad \text{gap } 0.22\%$$

If exact: $\varepsilon_0 = \alpha/(n(n+1)\sqrt{\varphi})$ — zero free parameters. Status: Hypothesis. Not yet tight enough for K-series.

Document Status

Document	Version	Key content
CRF Paper v6	current	$\mathcal{G}$, K35–K38, Kerr
CRF ThreePhase	v2	3-phase; step fn; f_{II} derived
CRF ThetaTower	v2	Tower; 3 interpretations
CRF BetaGamma	new	$\beta = d_s \mathcal{G}$; inflation endpoint
CRF MatterSplit	new	$n = d_s + n_{\text{m}}$; powerset; Hodge
CRF FormStructure	new	K35 uniqueness; proves Axiom 0
CRF StateSummary	v19	This document

Open Items

1. $\alpha\sqrt{\varphi} = D_0^2$ **exact?** If so, $\varepsilon_0 = \alpha/(72\sqrt{\varphi})$ and CRF has zero free parameters. Tighten gap below 0.05%.
2. **Inside matter types:** How do the 5 archetypes (1+3+1) generate specific particles? What determines their mass hierarchy?
3. **Loop correction:** Formal derivation of $1/((n-1)n_{\text{m}})$ from Psi-cluster geometry.
4. **P0-H Subject 4 (bracketed):** Empirical gate. Flat 50% prediction. $n \geq 50$ trials.

Transfer to next session

CRF has **zero free structural parameters**. The structure ($d_s = 3$, binary δ , $n = 8$, $n_{\text{m}} = 5$, 5 matter types 1+3+1, Hodge duality of space and gauge) follows from one algebraic theorem (K35 form uniqueness) alone.

CRF has **one free quantitative parameter**: ε_0 (or w_{DE}). DESI 2024 measures $w_{\text{DE}} = -0.754$, giving $\varepsilon_0 = 0.007565$ (CRF value 0.00755, gap 0.19%). These are the same degree of freedom.

If $\alpha\sqrt{\varphi} = D_0^2$ (gap 0.22%) is proved exact, even this last parameter is derived: $\varepsilon_0 = \alpha/(72\sqrt{\varphi})$. CRF would then have *zero* free parameters at any level.

The universe has three dimensions, operates in binary, has eight modes, five matter archetypes, and one dark energy equation of state.

None of these are free choices.

They are the unique solution to the requirement that the coupling constant equals the total complexity of all the forms that space can carry.

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